

SIRIVANTH SAI BANDREDDI

Fremont, California, USA | (+1) 405-986-1475 | [Linkedin: Sirivanth sai](#)

PROFESSIONAL SUMMARY

Data Scientist and AI/ML Engineer with 6+ years of combined experience spanning **data quality engineering**, **machine learning model design**, **NLP pipeline development**, and **business analytics**. Currently deployed at **Meta** validating largescale **AI/ML datasets** under Regulatory and Response Operations. Proven track record designing **ETL pipelines**, building **supervised and unsupervised ML models**, developing **NLP-based automation systems**, and delivering actionable insights through **Power BI** and **Tableau**. Proficient in **Python**, **R**, **SQL**, **TensorFlow**, **PyTorch**, **Scikit-learn**, **AWS**, and **Snowflake**. Adept at translating complex analytical findings into clear, actionable insights for both technical and non-technical stakeholders, with a strong track record of delivering measurable improvements in pipeline quality, model performance, and business outcomes.

TECHNICAL SKILLS

Data Analysis: **SQL** (Snowflake, MySQL, MSSQL, BigQuery), **Python** (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), **R** (ggplot2, dplyr), **Excel** (Pivot Tables, VLOOKUP, ad-hoc analysis)

Visualization: **Tableau**, **Power BI**, **Looker**, **Grafana**, **ArcGIS**, **QGIS** - dashboard development, KPI reporting, geospatial mapping

ML & Modeling: **Regression**, **Classification**, **Clustering** (K-Means), **Random Forest**, **AdaBoost**, **XGBoost**, **SVM**, **Deep Learning** (TensorFlow, PyTorch), **SARIMA**, **Quantitative Statistics**, **A/B Testing**, **Hypothesis Testing**, **Sentiment Analysis**, **NLP** (NLTK, SpaCy, Tokenization), **Foundation Models**, **DQN**

AI Automation & Testing: **Playwright**, **Selenium**, **Postman**, **REST API Testing**, **QA Framework Automation**, **ML Assisted Anomaly Detection**, **OCR Pipeline QA**, **Data Validation**, **Defect Analysis**, **Bug Triage**, **Schema Validation**, **MLOps Governance**, **Inter-Rater Reliability Calibration**

AI Tools: **OpenAI API** (Codex), **Claude Code** (Anthropic), **Gemini 3.5**, **LangChain**, **Hugging Face Transformers**

Data Engineering: **ETL Pipelines**, **Google Cloud Dataflow**, **Apache Hive**, **SAP HANA**, **Data Wrangling**, **Data Governance**, **Data Warehousing**

Cloud & Platforms: **AWS** (EC2, Lambda), **Snowflake**, **Google Cloud Platform**, **Microsoft Azure**, **Salesforce CRM**, **SAP HANA**

Project Management: **Agile** (Scrum, Kanban, Lean), **JIRA**, **Atlassian**, **Confluence**, **Stakeholder Communication**, **Cross functional Collaboration**, **SLA**, **Adherence**

PROFESSIONAL EXPERIENCE

Data QA Engineer

Sep 2025 – Present

Meta — Regulatory & Response Operations | Fremont, CA (On-site)

- Designed and deployed scalable **ML assisted anomaly detection** workflows validating large-scale AI/ML training datasets within the **WhatsApp Testing pipeline**, achieving **94% defect detection precision** and reducing false positives by 30% across 1,000+ files validated weekly against defined SLAs.
- Built and maintained **Playwright** and **Selenium** based automated test suites for **AI integrated WhatsApp features**, enabling continuous regression coverage across Android and web platforms and **reducing manual testing** effort by 35%.
- Developed **REST API validation** scripts using **Postman** and **Python** to verify AI/ML data pipeline endpoints, ensuring payload integrity, schema compliance, and response accuracy across **Snowflake** backed services.
- Authored and maintained **XML based test configuration files** and data schemas to standardize **QA workflows** and drive consistency across automated validation pipelines for AI/ML datasets.
- Built Python based **OCR integrated** data validation pipelines using **Pandas** and **NumPy** for structured and unstructured dataset QA, including **image embedded text extraction** by reducing manual validation effort by 40% and improving pipeline throughput by 25%.
- Applied advanced **SQL** and **Snowflake** to audit structured and semi-structured AI/ML datasets at scale, identifying and resolving **150+ schema inconsistencies** monthly, maintaining **98% dataset accuracy** and integrity for downstream model consumption.
- Conducted **root cause analysis** and causal reasoning-driven defect identification, reducing defect recurrence rate by 35% through inter-rater reliability calibration and structured QA reporting aligned with **MLOps governance standards**.
- Developed semi-automated QA validation frameworks integrating Python scripting with Snowflake-backed pipelines, cutting average defect resolution time from 48 hours to 18 hours and sustaining 96% SLA compliance across all WhatsApp Testing deliverables.

Graduate Teaching Assistant (Tutor) & ML Researcher – MS in Business Analytics
University of the Pacific | Eberhardt School of Business | Stockton, California, USA

August 2024 – May 2025

- Directed applied ML research and development of an **NLP-based feedback automation system** for **Instagram Threads App**, achieving **88.3% text classification accuracy** and reducing manual review workloads by 40% benchmarking **SVM, KNN, Decision Tree, Random Forest**, and **AdaBoost models** using **Python (NLTK, Pandas, Seaborn)** and **RapidMiner** across **precision, recall, and F1 metrics**.
- Designed and implemented a **Snowflake-compatible MSSQL database management system** with **15+ optimized entities** via BigQuery schema design, achieving **99% data accuracy**, **50% faster resource tracking**, and 70% improved query performance demonstrating Snowflake aligned data warehousing best practices.
- Delivered hands on instruction across all **MSBA program courses** in **Python, R, SQL (Snowflake, MSSQL, Grafana)**, **statistical modeling**, and **supervised learning** by mentoring Teaching Assistants and translating complex quantitative concepts into clear, accessible narratives for graduate cohorts.
- Applied quantitative research methods including **hypothesis testing, A/B testing, sentiment analysis**, and data querying to graduate ML coursework, reinforcing survey data analysis and insight communication skills across diverse audiences.

Information Technology Business Analyst
Asian Technologies | Hyderabad, India

October 2022 – July 2023

- Spearheaded strategic data analysis and **business intelligence** for **enterprise ERP products (SoftChits, MLMS)** spanning India, UAE, and USA, supporting **\$1.75M+** in business growth and **93% stakeholder satisfaction** across cross-regional teams.
- Streamlined **financial data operations** using **MSSQL, Apache Hive, Snowflake-compatible SQL pipelines, Python, Tableau**, and **Power BI** enabling 15% faster processing, 65% quicker reporting, and 30% faster delivery through Agile execution.
- Conducted comprehensive **gap analysis (AS-IS/TO-BE)** on \$13.5M+ enterprise products, translating findings into quantitative business cases and executive presentations for cross-regional, multinational stakeholders.
- Achieved 18% higher development efficiency and reduced requirement clarification cycles by 40% through structured stakeholder collaboration across cross-functional teams in India, UAE, and the USA.

Data Analyst
Asian Technologies | Hyderabad, India

March 2022 – September 2022

- Designed and implemented ETL pipelines using **Google Cloud Dataflow, SAP HANA, Snowflake, Python**, and **SQL** processing **2,000+ records** with 99.2% accuracy, achieving **45% faster queries** and 35% improved data efficiency.
- Developed **Tableau and Power BI dashboards** for **pharmaceutical and retail clients**, reducing decision-making time by 28% across UAE, India, and USA markets through compelling, **narrative-driven data visualizations**.
- Automated Snowflake-integrated reporting workflows using **Python (Pandas, NumPy, Seaborn)** and **R (dplyr, ggplot2)**, **saving 15+ hours weekly** and enabling 22% client growth through optimized data delivery.
- Supported **international partners** across India, Dubai, USA, and UK on ERP, CRM, Snowflake, and **cloud data strategies**, directly contributing to **\$1M+** in revenue growth through insight-driven recommendations.

KEY PROJECTS

Graduate:

U.S. Border Crossing Operations — Data-Driven Analytics (Graduate Research Project)

- Analyzed **396,401 entries** from **data.gov** applying **SARIMA modeling (AIC: 10,495.12)** for **traffic forecasting**, Random Forest Regressor (**MSE: 2.26B**), data analysis and model design using **Python (Jupyter Notebook, VS Code)**, K-means clustering to optimize resource allocation across 3 distinct port categories and created **3D geospatial visualizations** and **predictive models** in **3D geospatial visualizations** and **predictive models** in **Power BI** uncovering **300K+ US-Canada** and **100K+ US-Mexico** border entry patterns.

Urban Happiness Index Prediction — Deep Learning for Global City Planning

- Developed a **Deep Q-Network (DQN) model (PIYAAI_2)** with 93% accuracy to **predict happiness scores** for 136 cities, projecting a 15% increase in urban satisfaction and 40% reduction in decision-making timelines using **Python, TensorFlow, PyTorch, SPSS, Tableau, ArcGIS**, and **QGIS**.
- Identified **key correlations** ($R^2 = 0.87$ between green spaces and happiness), enabling data-driven healthcare and environmental policy recommendations for urban planners.

Predictive Pricing Model for Round-Shaped Diamonds (0.70–0.89 Carat)

- Built a **regression model** in **Microsoft Excel** to **forecast diamond prices** based on KPIs including carat weight, cut, color, and clarity using data from Blue Nile, With Clarity, James Allen, and Brilliant Earth.
- Analyzed and visualized data using **Pivot Tables**, **heat maps**, and scatter charts to compare natural versus lab-grown diamond prices and identify attribute-price relationships.

Music Streaming Analytics Dashboard - SQL, BigQuery & Grafana

- Built an advanced analytics dashboard using **SQL**, **Google BigQuery**, and **Grafana** to analyze **1,500+ songs (1901–2014)** across **sales**, **streams**, **downloads**, and **radio plays** from Spotify and delivered interactive **KPI visualizations** revealing total sales growth to \$3.569B and downloads reaching \$1.077B from 1999 to 2014.

Stockton Symphony Concert Analysis System

- Designed an **MSSQL** Server database with an ERD of **4 interconnected entities** to manage **2,800+ ticket transactions** from raw data and enhanced **revenue forecasting** accuracy by 40%, drove a 25% increase in ticket sales, and reduced analysis time by 60% through **automated reporting tools**.

Top 50 U.S. Tech Companies — Sector & Market Research Analysis

- Directed a comprehensive **data science project** analyzing trends across **7 major tech sectors** using **Python (Pandas, Matplotlib, Seaborn, Plotly)** by achieving 92% data normalization accuracy and built interactive **3D scatter plots** and **radar charts** uncovering **performance correlations** across **revenue**, **market cap**, and employment metrics for **Fortune 500 companies**.

EDUCATION

Master of Science in Business Analytics GPA: 3.87	Aug 2023 - Dec 2024
<i>University of the Pacific, (Stockton, CA, USA)</i>	
Bachelor of Technology in Electronic and Communication Engineering	Aug 2018 - July 2022
<i>Malla Reddy College of Engineering and Technology, (Hyderabad, TS, India)</i>	

CERTIFICATIONS

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| • AWS Certified Cloud Practitioner | • Google Data Analytics Professional |
| • Microsoft Power BI Data Analyst (PL-300) | • Tableau Desktop Specialist |
| • Microsoft Azure Data Fundamentals (DP-900) | |